Application of Multimodal for Lithology-aware Prompt Engineering in Mineral Processing Systems

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Coal and gangue are two important factors in the coal mining, and separating them from each other is one of the steps in the coal mining engineering. Because of the transportations and personnel fee, coal mining engineers should focus on separating coal and gangue from each other as much as possible to save the corresponding cost. Traditional coal mining industry is high-risky and labor-intensive, and simultaneously, until now the number of young professional mining engineers is still decreasing. The mentioned challenges are driving the government to transfer the coal mining industry from human-centered (human eyes locating, human hands taking) to automatic-robots-industry-centered (sensor/camera locating, robots arms taking). Up to now, combined with the robot arms, there are several computer vision algorithms (YOLO, RCNN) trained with open-source public dataset, and applied in the practical coal mining projects. During these projects, there are still some tasks that have not been solved already, i.e., to some degree, open-source public dataset cannot cover all the practical conditions. Therefore, discovering an approach of increasing the diversity of the existing dataset is in need. Based on the aforementioned issue, this research proposes to prove the possibility of applying the Generative Artificial Intelligence (AI) as the supplementary of the open-source public dataset (i.e., generating gangue images). Generative AI-based dataset mainly includes 2 patterns, individually: txt2img and img2img. With the assistance of the multimodal, the possibility of applying lithology-aware prompt engineering in generating gangue images has been proved. And the authors have categorized and analyzed the lithology-aware prompts.

Key Words: *AIGC, Prompt Engineering, Mining, Multimodal, YOLOv8*

1. INTRODUCTION

The coal mining industry faces significant challenges in efficiently separating coal from gangue (non-combustible rock), a critical step for cost reduction and operational safety[1), 2)]. Traditional methods, reliant on labor-intensive processes and human expertise, struggle with declining skilled workforce availability and high risks. To address this, the industry is transitioning toward automation, leveraging computer vision algorithms like You Only Look Once (YOLO) trained on open-source datasets[3)]. However, these models often lack robustness in diverse real-world conditions due to limited dataset diversity.

Artificial Intelligence Generated Content (AIGC), particularly through tools like Stable Diffusion (SD), offers an approach to enhance dataset diversity [4), 5)]. By generating synthetic images via two primary modes—txt2img (text-to-image generation using keyword prompts) and img2img (image-guided generation combining background references and prompts), AIGC can create supplementary training data that mimics complex industrial scenarios[6)].

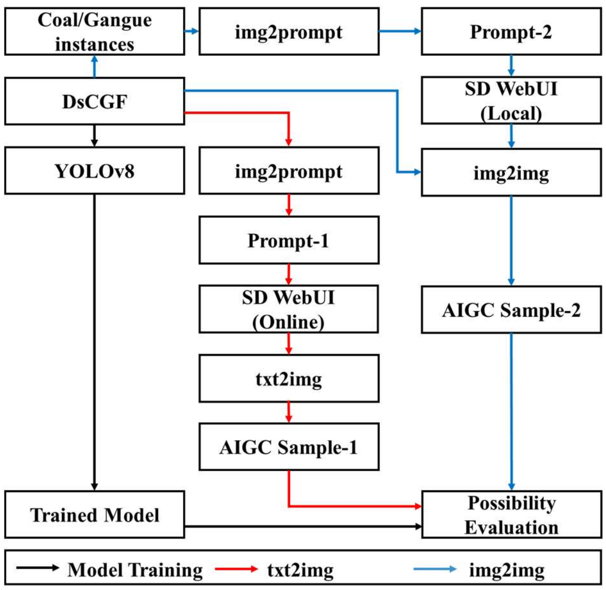


Fig.1 Workflow of how to apply the prompt-assisted AIGC Dataset on the coal and gangue-related object detection.

As shown in **Fig.1**, the authors showed a workflow that how to apply the prompt-assisted AIGC Dataset on detecting the coal and gangue. The main focus of this research is based on the possibility of using prompt engineering to simulate the gangues in the complex industrial scenarios (conveyor belt mining transferring system). A trained YOLOv8 coal and gangue object detection model was used to test the possibility.

2. METHODS

(1) Model Training

Fig.2 is a sample derived from an open-source dataset called DsCGF (A large-scale open image dataset for deep learning enabled intelligent sorting and analyzing of raw coal with CC 4.0 license) which includes both coal and gangue objects[2)].

Among all the parts in DsCGF, "ERLINTU" part derived from Inner Mongolia Erlintu Coal Mine was chosen as the dataset for input. This selected ERLINTU part is with a high-angle close-up security camera view, which can record the movement of coal and gangue under productive state with limited light condition (considering of the underground safety). As shown in **Fig.3**, the authors applied YOLOv8 model to train the ERLINTU part dataset with around 35,000 coal and 5,000 gangue instances. The ratio of coal and gangue was unbalanced, and this is the main reason why the authors mimic the gangues in the conveyor belt mining transferring system. The trained YOLOv8 model will be the tool for evaluating the possibility of applying the generated gangue in the practical data augmentation or not.

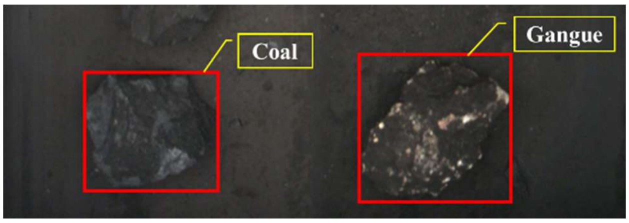


Fig.2 This is a sample image selected from DsCGF dataset ERLINTU part.

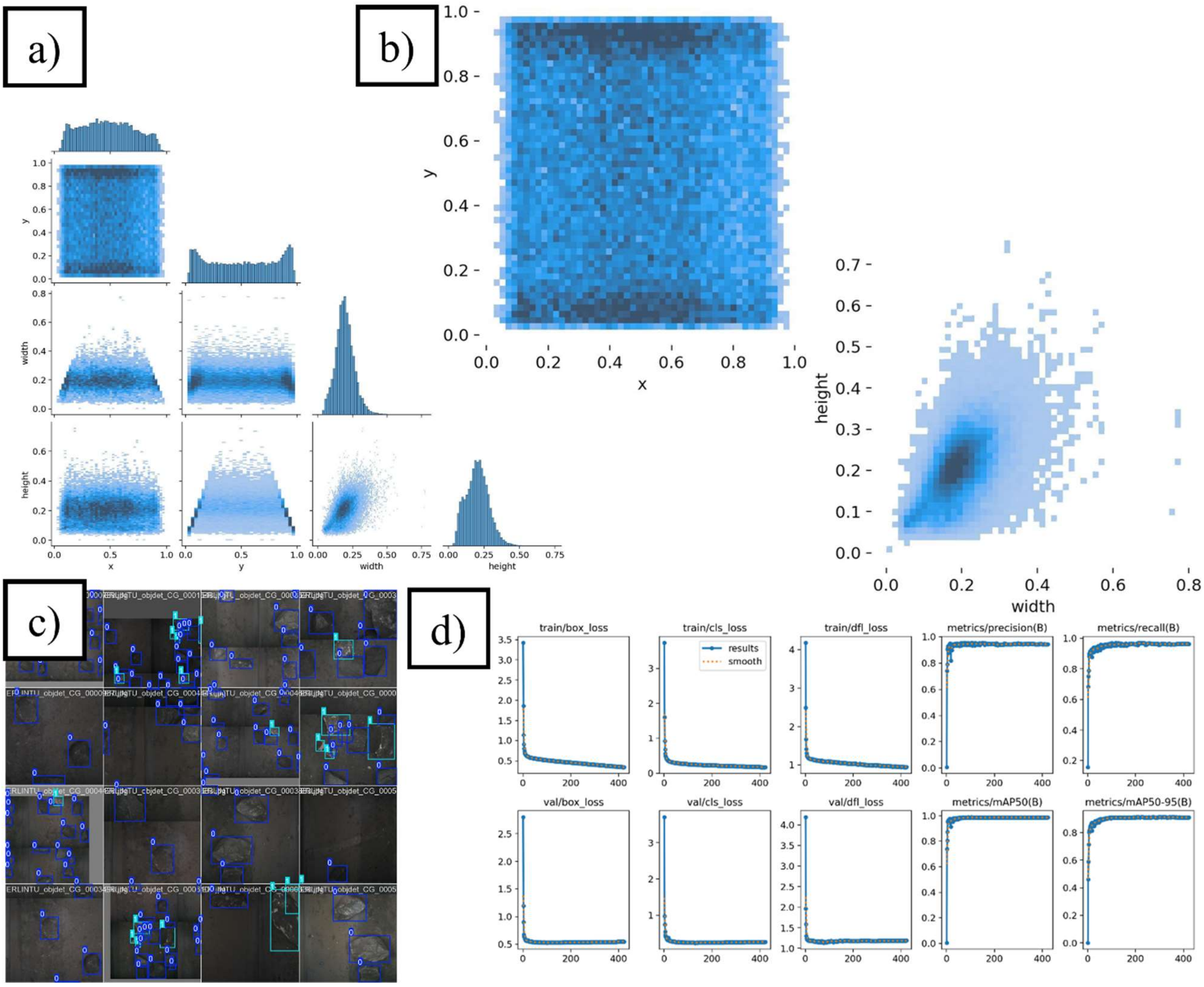
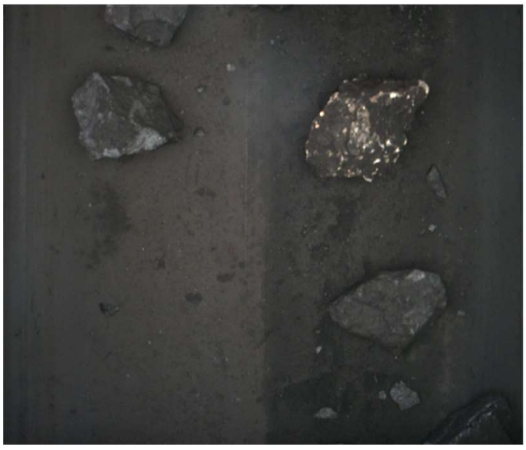


Fig.3 DsCGF ERLINTU-part YOLOv8 object detection model training process.

Original Image (2275 pix × 1925 pix)	Prompts



Prompt:
"A minimalist still life composition of rough-textured rocks scattered randomly across a dark surface, irregular shapes with varied geological textures and color tones (dark gray, charcoal, slate), slight reflective surfaces catching subtle light, coarse grainy background with weathered stains and natural imperfections, muted color palette, dramatic shadow casting, cold industrial atmosphere, high detail mineralogical study, macro photography style, soft directional lighting"
Texture descriptors: "rough-textured", "coarse grainy", "weathered stains"
Geological details: "varied geological textures", "mineralogical study"
Lighting: "soft directional lighting", "dramatic shadow casting"
Color constraints: "muted color palette", "dark gray, charcoal, slate"
Composition: "scattered randomly", "minimalist still life"
Photo realism cues: "macro photography style", "high detail"
Negative prompt:
"blurry, smooth surfaces, vibrant colors, vegetation, human figures, symmetrical arrangements, cartoonish style, digital painting"

Fig.4 A sample image selected from DsCGF (left) and img2prompt-based prompts, which include positive and negative ones (right).

Fig.5 Image generation derived from Stable Diffusion WebUI online version (left: coal, right: gangue)

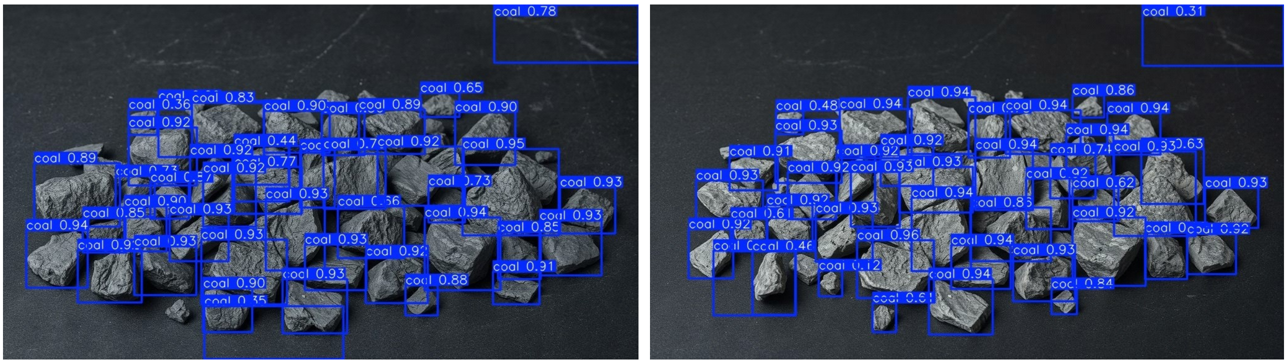


Fig.6 Result of the objects detected by the trained YOLOv8 model (left: coal, right: gangue), no gangues have been detected as the gangues.

(2) txt2img Process

As shown in **Fig.4 (left-side)**, the authors chose another sample image as the input for img2prompt function to generate the prompts that can be used in SD WebUI online version[7]. **Fig.S1** showed the detailed processing of how prompts were generated. In this research, the img2prompt function was mainly based on the Tencent-YuanBao[8]-interfaced large language model (LLM) DeepSeek R1. The generated prompts include both positive and negative ones, i.e., Prompt and Negative prompt. Prompt part includes not only comprehensive descriptions (surface, light, atmosphere and style), also the details of the individual objects (textures). Negative prompt includes the words that can mislead the image generation.

After referring the prompts in **Fig.4**, **Fig.5** showed the image generations of coal and gangue, individually. As shown in **Fig.6**, no gangues have been detected as the gangues using the trained YOLOv8 model. Because of the limitations derived from the mentioned textures-related prompts, SD was not able to understand the "feature" of the gangue that was considerate as the main reason of the misclassification between coal and gangue. And too much less-relative describable prompts can reduce the concentration of SD.

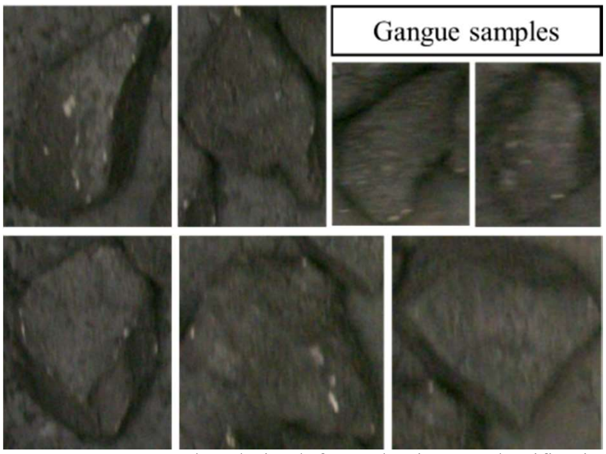


Fig.7 Gangue samples derived from the image classification results in ERLINTU part.

Table 1 Prompts (DeepSeek) used for the conversation between user and LMM with different purposes.

Specification	Details
Prompt-related	I want to generate an image that can include the uploaded images. I will use stable diffusion to generate it, and I need prompt that can describe these figures. Please show me the prompt. I am using sd-v1-4.ckpt [fe4efff1e1].
Summary-related	Please summarize the prompt related to the gangue based on the mentioned content with strategy

Table 2 Key-feature-related prompts derived from shape and lithology-related categories.

Specification	Details
Shape-related	jagged edges, shield-shaped geological formation oxidized cracks volcanic glass texture rough charcoal-black surface
Lithology-related	white crystalline speckles dendritic oxidation patterns granular textures ash tones
Negative-related	human figures color images clean surfaces cartoonish style Vegetation water droplets symmetrical patterns text descriptions coal dust clouds 3D render

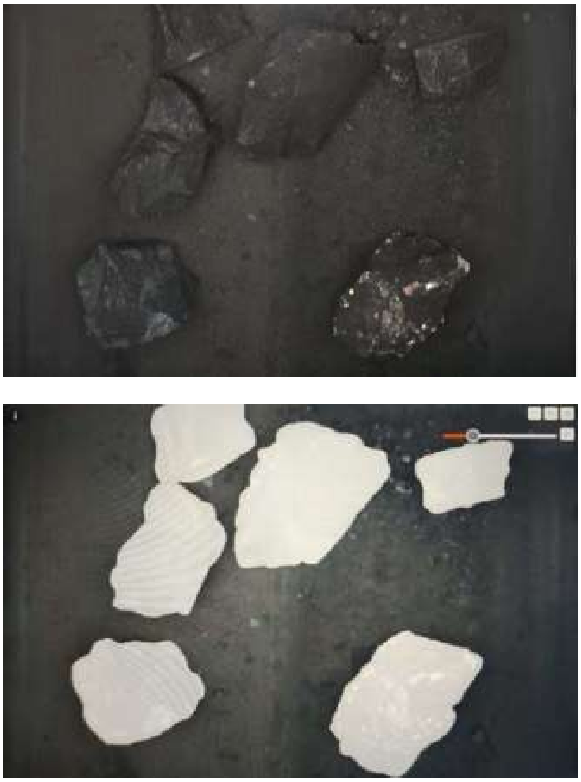


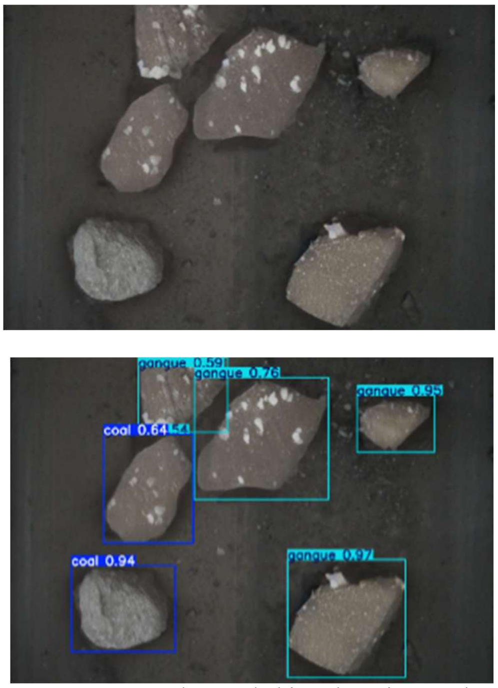
Fig.8 Original image and corresponding Inpainting mask (upper-side: original image; down-side: inpainting mask, white).

Table 3 Versions of the software and hardware used for Stable Diffusion installation.

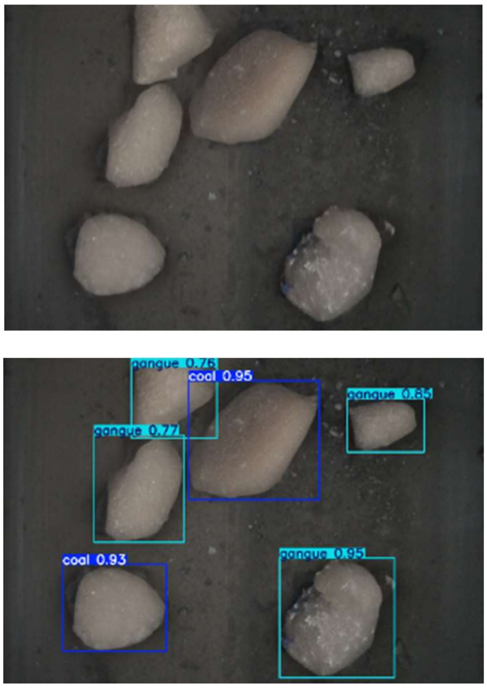
Specification	Details
OS	Ubuntu 22.04.4 LTS
CUDA Version	12.4
NVIDIA-SMI	550.144.03
GPU	NVIDIA GeForce RTX 3090

Table 4 Parameter setting of the Generation-Inpaint module in the Stable Diffusion WebUI local version and corresponding chosen options.

Specifications	Details
Resize mode	Just resize (default)
Mask blur	64 (maximum, default=4)
Mask mode	Inpaint masked (default)
Masked content	latent nothing
Inpaint area	Only masked
Only masked padding, pixels	256 (maximum value)
Sampling method	DPM++ 2M
Schedule type	Automatic
Sampling steps	150
Width/Height	608/520
Batch count	100 (maximum, default=1)
Batch size	8 (maximum, default=1)
CFG Scale	7 (default)
Denoising strength	0.75 (default)



(a) AIGC sample-1 and object detection result.



(b) AIGC sample-2 and object detection result.

Fig.9 AIGC and corresponding object detection results.

(3) Img2img process

As shown in **Fig.6**, there were no generated gangue items. Correspondingly, "how to describe the gangue items in logical and scientific way that SD can understand" should be in consideration. As depicted in **Fig.7** and **Table 1**, the authors used gangue samples and prompts (DeepSeek) for conversation in img2prompt function module to generate the prompts (Stable Diffusion WebUI, local version with Inpaint function). As shown in **Table 2**, the authors categorized the key-feature-related prompts into 2 types, i.e., shape and lithology, which was derived from the response in **Fig.S1**. **Fig.8** showed one mask inpainting sample with white color.

During the process of the img2img mask inpainting, not only the mask-related operations, there are also several parameters that should be set. As shown in **Table 3 and 4**, all the details of the necessary specification for local version SD are displayed.

3. RESULTS AND DISCUSSIONS

(1) Object Detection Results

As shown in **Fig.9**, there are two AIGC image samples and corresponding object detection results. In **Fig.9**, even the prompts for individual instance were used as same, not all the instances have been detected as gangue. To understand the common features of the surface texture, individual consideration on each instance is necessary.

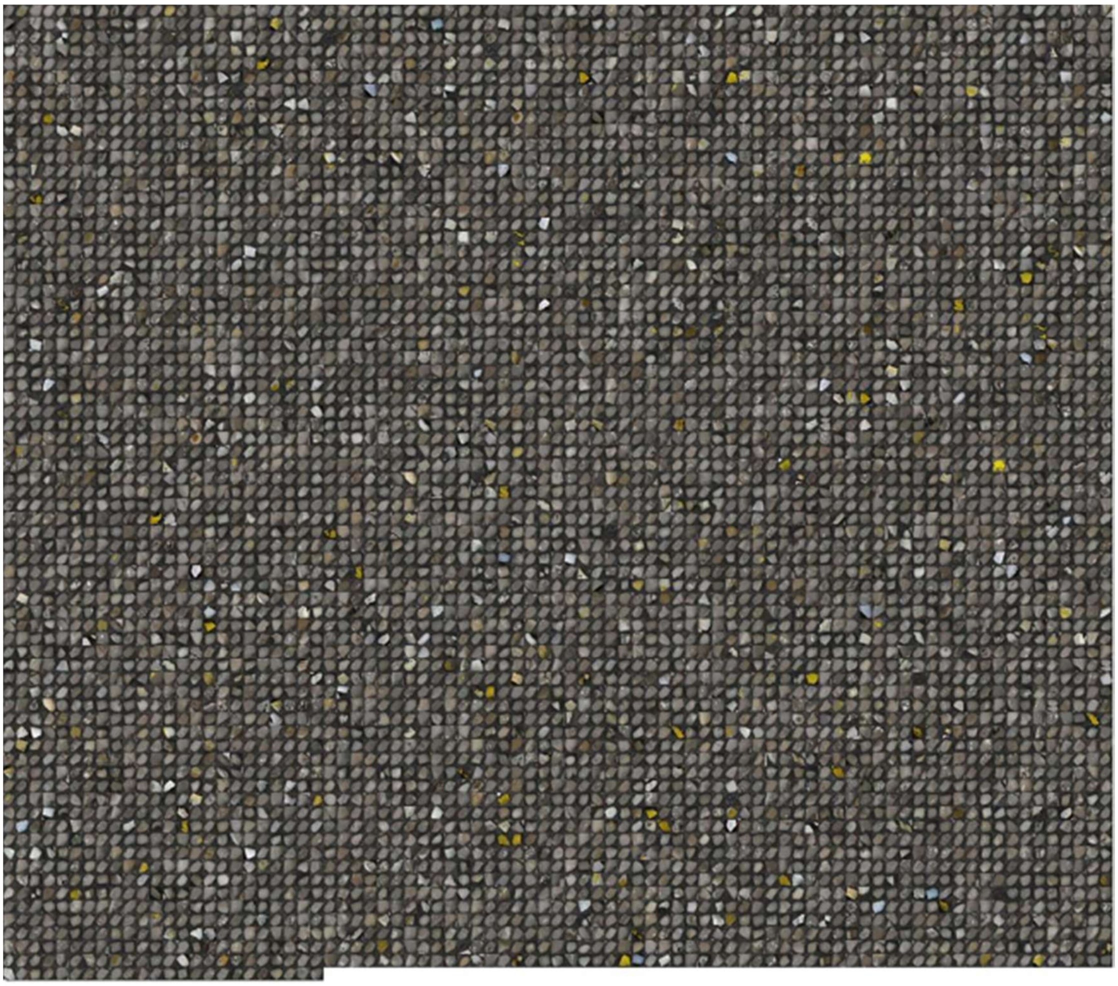
(2) Mosaic mixture

Considering of the reasons why some AIGC images can be detected, the authors should understand what kind of "feature" are useful for YOLOv8 object detection. For easier understanding, **Fig.10** showed two mosaic mixture images derived from original and AIGC gangue instances. Obviously, original gangue instances have more diversity, and the light condition or brightness are varying in different instances.

(3) Heatmap

Fig.11 showed the heatmap and overlay images of the gangue instances, there are some common heat parts (red part) between original and AIGC that can be considerate as useful "feature" for YOLOv8 object detection. Correspondingly, the prompts are proved successfully to be re-generated. The authors visualized the layer -2 (Deep Layers) for reflecting high-level semantic features, which includes but not limited to Object components (e.g., mineral clusters, metallic reflective zones), Contextual dependencies (e.g., ore-background transitions), Geological anomalies (e.g., oxidized regions with unique color/texture). And these mentioned three factors are most important factors which lead to the segmentation of gangue and coal. Because the background is transferring belt and light condition is dark, "Object components" and "Contextual dependencies" are not the object factors in this research (dark light and simple background). As conclusion, Geological anomalies are much more important than other two factors. This is also the reason why red-color of the generated Gangue heat-map in **Fig.11** concentrated in the middle area.

(a) Mosaic mixture image derived from 1360 original gangue instances (13 px × 13 px per instance image).



(b) Mosaic mixture image derived from 6000 AIGC gangue instances (13 px × 13 px per instance image).

Fig.10 Mosaic mixture images derived from original and AIGC gangue instances.

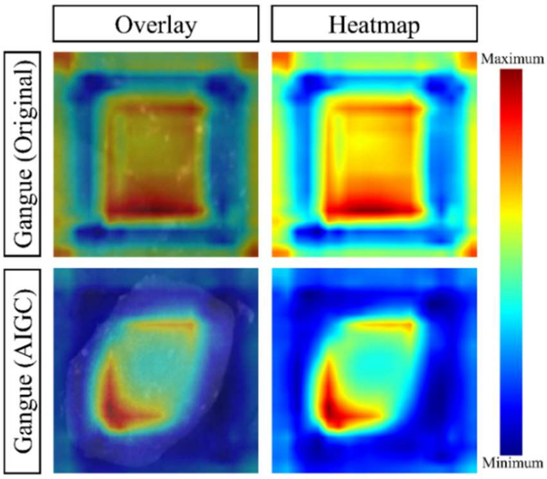
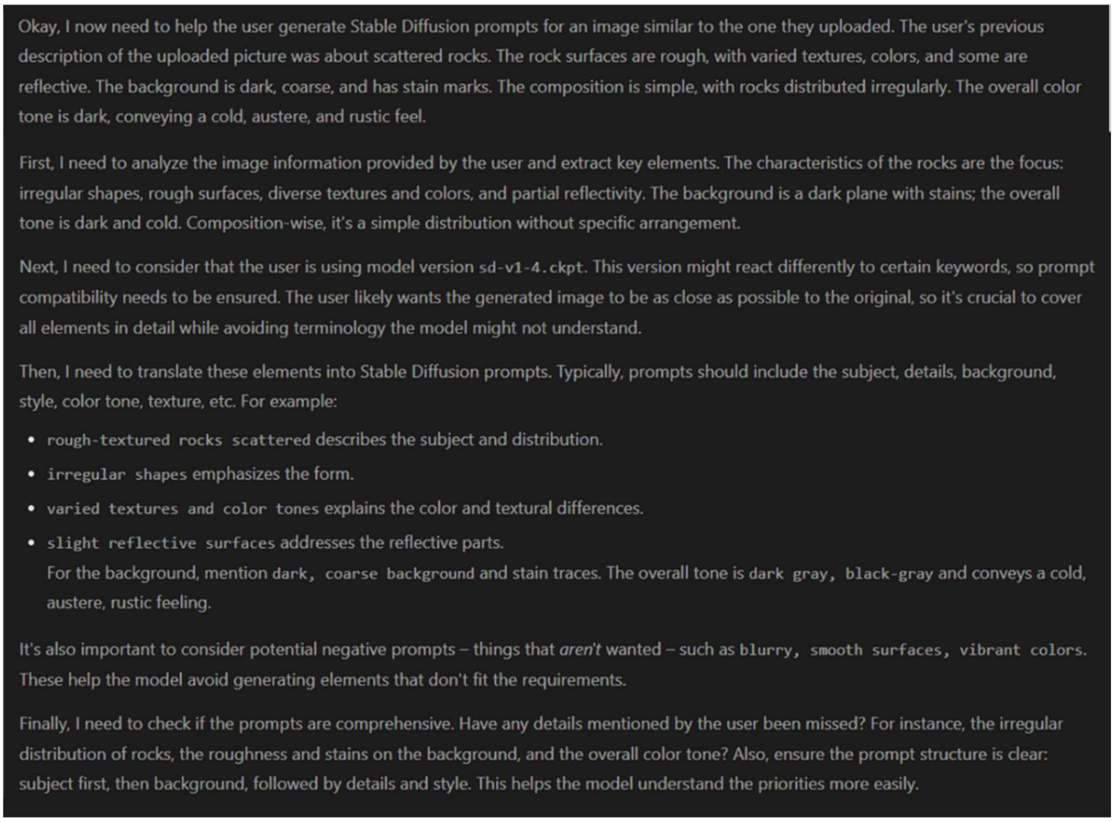


Fig.11 Heatmap and overlay images of all the gangue instances, original and AIGC, individually.



Okay, I now need to help the user generate Stable Diffusion prompts for an image similar to the one they uploaded. The user's previous description of the uploaded picture was about scattered rocks. The rock surfaces are rough, with varied textures, colors, and some are reflective. The background is dark, coarse, and has stain marks. The composition is simple, with rocks distributed irregularly. The overall color tone is dark, conveying a cold, austere, and rustic feel.

First, I need to analyze the image information provided by the user and extract key elements. The characteristics of the rocks are the focus: irregular shapes, rough surfaces, diverse textures and colors, and partial reflectivity. The background is a dark plane with stains; the overall tone is dark and cold. Composition-wise, it's a simple distribution without specific arrangement.

Next, I need to consider that the user is using model version `sd-v1-4.ckpt`. This version might react differently to certain keywords, so prompt compatibility needs to be ensured. The user likely wants the generated image to be as close as possible to the original, so it's crucial to cover all elements in detail while avoiding terminology the model might not understand.

Then, I need to translate these elements into Stable Diffusion prompts. Typically, prompts should include the subject, details, background, style, color tone, texture, etc. For example:

- `rough-textured rocks scattered` describes the subject and distribution.
- `irregular shapes` emphasizes the form.
- `varied textures and color tones` explains the color and textural differences.
- `slight reflective surfaces` addresses the reflective parts.
  For the background, mention `dark, coarse background` and stain traces. The overall tone is `dark gray, black-gray` and conveys a cold, austere, rustic feeling.

It's also important to consider potential negative prompts – things that *aren't* wanted – such as `blurry, smooth surfaces, vibrant colors`. These help the model avoid generating elements that don't fit the requirements.

Finally, I need to check if the prompts are comprehensive. Have any details mentioned by the user been missed? For instance, the irregular distribution of rocks, the roughness and stains on the background, and the overall color tone? Also, ensure the prompt structure is clear: subject first, then background, followed by details and style. This helps the model understand the priorities more easily.

Fig.S1 This is a sample image of thinking process under deep-think mode using Tencent-YuanBao-interfaced DeepSeek R1. During this process, the LLM should understand the purpose derived from the requirements clearly. Then the LLM will analyze how to translate the natural language into the prompts that Stable Diffusion can understand, use and generate the images correspondingly. Except of the prompts, negative prompts should also be considerate. Finally, the LLM summarized the answers and commented. The characteristics of the rocks are the focus, i.e., irregular shapes, rough surfaces, diverse textures and colors, and partial reflectivity. Table 2 was derived from these rock-characters-focused factors.

4. CONCLUSION

This study presented a novel method for AI-based gangue image generation, using multimodal-related prompt engineering. The generated lithology-aware gangue data augmentation have been proved possibly applying in mineral processing systems that was tested by trained YOLOv8 gangue detector. Comparing with the whole-image-derived prompts, the instances-derived prompts and specific area inpainting can support to generate the images more productively. Systematic categories-based prompts can increase the possibility of generating the targeted objects and background.

Tencent-YuanBao-interfaced, LLM, DeepSeek R1 showed its availability of generating the prompts derived from several (10) instances images, its summary and categorizing ability on the prompts.

5. FUTURE WORKS

The gangue-related images have been generated as the data augmentation for the mineral processing, reducing the difficulty and burden of collecting the data on-site. But, the object area selection in mask inpainting operation also need personnel to handle, which reduce the productive speed. Some segmentation models, like Segment Anything Model (SAM), should be considerate to include in the workflow in the near future to reduce the personnel operation.

Especially when coal and gangue have common features with each other, the researchers should use 2-Step method (i.e., Step-1, segmentation using SAM or instance segmentation model to extract from background; Step-2, classification using YOLOv8-cls to individual objects into coal and gangue) to extract them from the background and each other clearly. And in the Step-2, the developed and validated prompts can be used for training the classification model.

The prompts, LLM and LMM (Large Multimodal Model) have been already applied in multiple engineering and science scenes[9)-17)], including but not limited to biomedical, remote sensing, computer science. The manual and guidelines of specific backgrounds and wider application using prompts, LLM and LMM should be in consideration.

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